

Table S1. Virtual screening of the sixty designed compounds by the three-classifier GROVER ensemble.

Each of the sixty designed compounds (ten series, **I–X**, of six members each) was encoded with the fine-tuned GROVER representation and scored independently by three classifiers: a random forest (RF), XGBoost and a multilayer perceptron (MLP). For each classifier, *Active Probability* is the probability assigned to the Active class and *Predicted Label* applies a decision threshold of 0.5. *Active Hits* is the number of classifiers predicting Active (0–3); a compound was taken as predicted active when a majority, that is at least two of the three, agreed. Series **IX** is the only series in which all six members met this criterion, and was the series carried forward to synthesis.

Code	SMILES	RF		XGBoost		MLP		Active
		Prob.	Label	Prob.	Label	Prob.	Label	Hits
Series I								
Ia	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=C(C=CC=C4)C1)=O)=CC=C32)C=C1)NO</chem>	0.47	Inactive	0.29	Inactive	0.67	Active	1
Ib	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=CC(C1)=CC=C4)=O)=CC=C32)C=C1)NO</chem>	0.44	Inactive	0.32	Inactive	0.67	Active	1
Ic	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=CC=C(C=C4)F)=O)=CC=C32)C=C1)NO</chem>	0.26	Inactive	0.02	Inactive	0.27	Inactive	0
Id	<chem>O=C(CCC1=CC=C(CN2C3=CC(NC(C4=C(C=CC=C4)C1)=O)=CC=C3N=C2)C=C1)NO</chem>	0.47	Inactive	0.25	Inactive	0.67	Active	1
Ie	<chem>O=C(CCC1=CC=C(CN2C3=CC(NC(C4=CC(C1)=CC=C4)=O)=CC=C3N=C2)C=C1)NO</chem>	0.46	Inactive	0.16	Inactive	0.68	Active	1
If	<chem>O=C(CCC1=CC=C(CN2C3=CC(NC(C4=CC=C(C=C4)F)=O)=CC=C3N=C2)C=C1)NO</chem>	0.28	Inactive	0.03	Inactive	0.28	Inactive	0
Series II								
IIa	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=C(C=CC=C4)C1)=O)=CC=C32)C=C1)NO</chem>	0.29	Inactive	0.16	Inactive	0.21	Inactive	0
IIb	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=CC(C1)=CC=C4)=O)=CC=C32)C=C1)NO</chem>	0.37	Inactive	0.33	Inactive	0.61	Active	1
IIc	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=CC=C(C=C4)F)=O)=CC=C32)C=C1)NO</chem>	0.19	Inactive	0.01	Inactive	0.03	Inactive	0
IId	<chem>O=C(CCC1=CC=C(CN2C3=CC(CC(C4=C(C=CC=C4)C1)=O)=CC=C3N=C2)C=C1)NO</chem>	0.31	Inactive	0.16	Inactive	0.22	Inactive	0
IIe	<chem>O=C(CCC1=CC=C(CN2C3=CC(CC(C4=CC(C1)=CC=C4)=O)=CC=C3N=C2)C=C1)NO</chem>	0.37	Inactive	0.33	Inactive	0.62	Active	1
IIIf	<chem>O=C(CCC1=CC=C(CN2C3=CC(CC(C4=CC=C(C=C4)F)=O)=CC=C3N=C2)C=C1)NO</chem>	0.20	Inactive	0.01	Inactive	0.04	Inactive	0
Series III								
IIIa	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=C(C=CC=C4)C1)=CC=C32)C=C1)NO</chem>	0.36	Inactive	0.13	Inactive	0.19	Inactive	0
IIIb	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=CC(C1)=CC=C4)=CC=C32)C=C1)NO</chem>	0.44	Inactive	0.28	Inactive	0.46	Inactive	0
IIIc	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=CC=C(C=C4)F)=CC=C32)C=C1)NO</chem>	0.28	Inactive	0.01	Inactive	0.05	Inactive	0
IIId	<chem>O=C(CCC1=CC=C(CN2C3=CC(CCC4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)NO</chem>	0.36	Inactive	0.13	Inactive	0.19	Inactive	0
IIIe	<chem>O=C(CCC1=CC=C(CN2C3=CC(CCC4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)NO</chem>	0.43	Inactive	0.31	Inactive	0.50	Inactive	0
IIIIf	<chem>O=C(CCC1=CC=C(CN2C3=CC(CCC4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO</chem>	0.27	Inactive	0.03	Inactive	0.06	Inactive	0
Series IV								
IVa	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)NO</chem>	0.48	Inactive	0.40	Inactive	0.75	Active	1
IVb	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)NO</chem>	0.55	Active	0.86	Active	0.91	Active	3
IVc	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO</chem>	0.33	Inactive	0.01	Inactive	0.11	Inactive	0

continued on next page

Table S1 (continued)

Code	SMILES	RF		XGBoost		MLP		Active
		Prob.	Label	Prob.	Label	Prob.	Label	Hits
IVd	<chem>O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)NO</chem>	0.48	Inactive	0.24	Inactive	0.75	Active	1
IVe	<chem>O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)NO</chem>	0.57	Active	0.64	Active	0.90	Active	3
IVf	<chem>O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO</chem>	0.36	Inactive	0.03	Inactive	0.12	Inactive	0
<i>Series V</i>								
Va	<chem>O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)=O)NO</chem>	0.35	Inactive	0.25	Inactive	0.67	Active	1
Vb	<chem>O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)=O)NO</chem>	0.40	Inactive	0.15	Inactive	0.85	Active	1
Vc	<chem>O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO</chem>	0.27	Inactive	0.01	Inactive	0.09	Inactive	0
Vd	<chem>O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)=O)NO</chem>	0.36	Inactive	0.22	Inactive	0.69	Active	1
Ve	<chem>O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)=O)NO</chem>	0.41	Inactive	0.19	Inactive	0.86	Active	1
Vf	<chem>O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO</chem>	0.27	Inactive	0.04	Inactive	0.10	Inactive	0
<i>Series VI</i>								
VIa	<chem>O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)=O)NO</chem>	0.62	Active	0.88	Active	0.42	Inactive	2
VIb	<chem>O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)=O)NO</chem>	0.68	Active	0.88	Active	0.70	Active	3
VIc	<chem>O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO</chem>	0.42	Inactive	0.19	Inactive	0.03	Inactive	0
VIId	<chem>O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)=O)NO</chem>	0.63	Active	0.69	Active	0.42	Inactive	2
VIe	<chem>O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)=O)NO</chem>	0.68	Active	0.88	Active	0.69	Active	3
VIIf	<chem>O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO</chem>	0.41	Inactive	0.17	Inactive	0.03	Inactive	0
<i>Series VII</i>								
VIIa	<chem>O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)=O)NO</chem>	0.48	Inactive	0.17	Inactive	0.11	Inactive	0
VIIb	<chem>O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)=O)NO</chem>	0.49	Inactive	0.20	Inactive	0.28	Inactive	0
VIIc	<chem>O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO</chem>	0.32	Inactive	0.01	Inactive	0.01	Inactive	0
VIIId	<chem>O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)=O)NO</chem>	0.48	Inactive	0.17	Inactive	0.10	Inactive	0
VIIE	<chem>O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)=O)NO</chem>	0.51	Active	0.20	Inactive	0.26	Inactive	1
VIIIf	<chem>O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO</chem>	0.31	Inactive	0.02	Inactive	0.01	Inactive	0
<i>Series VIII</i>								
VIIIa	<chem>O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)NO</chem>	0.52	Active	0.60	Active	0.53	Active	3
VIIIb	<chem>O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)NO</chem>	0.52	Active	0.45	Inactive	0.78	Active	2
VIIIc	<chem>O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO</chem>	0.35	Inactive	0.01	Inactive	0.05	Inactive	0
VIIIId	<chem>O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)NO</chem>	0.50	Inactive	0.41	Inactive	0.53	Active	1
VIIIe	<chem>O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)NO</chem>	0.52	Active	0.45	Inactive	0.77	Active	2

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Table S1 (continued)

Code	SMILES	RF		XGBoost		MLP		Active
		Prob.	Label	Prob.	Label	Prob.	Label	Hits
VIII f	<chem>O=C(NCC1=CC=C(C(C(C)N2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO</chem>	0.33	Inactive	0.01	Inactive	0.05	Inactive	0
<i>Series IX</i>								
IXa	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=C(C1)C=CC=C4)=O)=CC=C32</chem>	0.60	Active	0.94	Active	0.96	Active	3
IXb	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=CC(C1)=CC=C4)=O)=CC=C32</chem>	0.57	Active	0.94	Active	0.96	Active	3
IXc	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=CC=C(F)C=C4)=O)=CC=C32</chem>	0.52	Active	0.31	Inactive	0.88	Active	2
IXd	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=C(C1)C=CC=C4)=O)=CC=C3N=C2</chem>	0.62	Active	0.95	Active	0.97	Active	3
IXe	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=CC(C1)=CC=C4)=O)=CC=C3N=C2</chem>	0.56	Active	0.93	Active	0.97	Active	3
IXf	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=CC=C(F)C=C4)=O)=CC=C3N=C2</chem>	0.51	Active	0.44	Inactive	0.89	Active	2
<i>Series X</i>								
Xa	<chem>O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)NO</chem>	0.58	Active	0.73	Active	0.78	Active	3
Xb	<chem>O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO</chem>	0.39	Inactive	0.09	Inactive	0.14	Inactive	0
Xc	<chem>O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)NO</chem>	0.60	Active	0.73	Active	0.78	Active	3
Xd	<chem>O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)NO</chem>	0.65	Active	0.94	Active	0.91	Active	3
Xe	<chem>O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO</chem>	0.40	Inactive	0.08	Inactive	0.14	Inactive	0
Xf	<chem>O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)NO</chem>	0.64	Active	0.95	Active	0.92	Active	3